

W007 — CI/CD Pipeline: Validation Report

Wave: W007 | **Status:** delivered | **Date:** 2026-06-07

This is the Normalized-Claim $\neq$ Complete record for W007 (per AGENTS.md governance): every claim below is backed by a file that exists and a runtime result that was actually produced.

1. Claim

The MINERVA P&ID pipeline is continuously verifiable: a GitHub Actions workflow regenerates all derived artifacts (`./make.sh`), runs the full standalone test battery under coverage, enforces a deterministic golden-file regression gate, checks clean-rebuild reproducibility, and uploads artifacts — all **without pytest**, honoring the project’s standalone-runner convention.

2. What exists (files)

Artifact	Path	Purpose
Workflow	<code>ci/minerva-pid-test.yml</code>	build + test + coverage + golden + reproducibility + artifacts (staged; see §6)
Coverage cfg	<code>.coveragerc</code> , [<code>tool.coverage.*</code>] in <code>pyproject.toml</code>	coverage.py settings (source=src, branch, parallel)
Golden snapshots	<code>tests/golden/expected_line_model_structure.json</code> , <code>.../expected_component_counts.json</code> , <code>.../expected_crossmap_stats.json</code>	pinned invariants from runtime output
Golden runner	<code>tests/test_golden_files.py</code>	standalone golden-file regression gate
Pre-commit	<code>.pre-commit-config.yaml</code>	advisory hygiene/lint hooks
Implementation doc	<code>docs/W007_CI_IMPLEMENTATION.md</code>	what was built + the no-pytest rationale
This report	<code>reports/W007_CICD_REPORT.md</code>	validation record

3. Runtime results (actually produced)

```
./make.sh                                [?] exit 0 (all derived outputs regenerated)
./make.sh --clean ; ./make.sh           [?] exit 0 (idempotent; artifacts re-created)

Standalone test battery (PYTHONPATH=src python3 tests/test_*.py):
test_colour_model.py                    5/5    PASS
test_golden_files.py                    4/4    PASS    (NEW this wave)
test_integration_pipeline.py            5/5    PASS
test_w003_w004.py                      10/10   PASS
test_w005_reconciliation.py            11/11   PASS
test_w006_crossmap.py                  13/13   PASS
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TOTAL                                  48/48   PASS    (6 runners)

Coverage (coverage run -p [?] combine [?] report [?] xml):
coverage.xml produced (~98 KB)
TOTAL line coverage [?] 17.9% (reported, not gated [?] see §5)
```

4. Golden-file gate — pinned values (verified)

Invariant	Value
<code>line_model.line_count</code>	9
<code>catalog unique_count / real_count / template_count</code>	165 / 141 / 24
<code>real_prefix_histogram</code>	CV 10, EH 22, HV 12, LS 25, PT 9, TT 63 (= 141)
<code>crossmap total_design_tags / total_asdrawn_real_tags</code>	97 / 141
<code>crossmap mapped (HIGH/MED/LOW)</code>	43 (0 / 39 / 4)
<code>crossmap unmapped_design / as-drawn_unclaimed</code>	54 / 98

Internal-consistency assertions enforced: histogram Σ = `real_count` ;
confidence tiers Σ = `mapped` . Volatile geometry and the ~1-byte XLSX jitter are
excluded from the gate by design → deterministic.

5. Known gaps

- **Coverage is reported, not gated.** ~18% reflects that tests target pure functions while `build_*` scripts execute via `make.sh` as `__main__` . No inflated `fail_under` was added. A real floor needs build-script importability refactoring — deferred.

- **Workflow cannot self-activate** — GitHub App lacks `workflows` permission; the file is staged at `ci/minerva-pid-test.yml` and a maintainer must copy it to `.github/workflows/` (instructions in §6 and the implementation doc).
- **Codecov upload commented out** (needs maintainer `CODECOV_TOKEN`).
- **pre-commit advisory only** — not enforced in CI to avoid a mass reformat diff that could perturb reproducible outputs.
- The wave-registry's broader W007 line also names cross-drawing QCELL↔RFCELL reconciliation and 4.5K/2K/warm segment classification; those are **modelling** scope deferred to a follow-up. This deliverable is the CI/CD half.

6. Maintainer activation (one-time)

```
cp MINERVA_PID/ci/minerva-pid-test.yml .github/workflows/minerva-pid-test.yml
git add .github/workflows/minerva-pid-test.yml
git commit -m "ci: activate MINERVA P&ID workflow (W007)"
git push
```

7. Reproduce locally

```
./make.sh
export PYTHONPATH=src
for t in tests/test_*.py; do python3 "$t"; done
# coverage:
for t in tests/test_*.py; do coverage run -p "$t"; done
coverage combine && coverage report && coverage xml
git checkout -- reports/W005_validation_report.md # revert XLSX byte jitter
```

pass = true — output files exist, this validation report exists, runtime counts are recorded above, and known gaps are listed.